

Spherical Functors and $\mathbb{P}^n$ -Functors

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Scope and source

This standalone note collects the categorical foundations in the live spherical-adjunction chapter and the independent $\mathbb{P}^n$ -functor appendix. It preserves the definitions, conventions, formal criteria, diagrams, proofs, and citations needed for a coherent reading. The derived-Grassmannian chapters and their applications are intentionally excluded. The Anno–Logvinenko autoequivalence theorem remains a cited theorem, as in the source appendix; its direct proof is not reproduced there.

Chapter 1

Spherical adjunctions and categorical twists

This chapter reviews spherical adjunctions, a standard source of nontrivial autoequivalences of derived categories and a categorical B-model counterpart of Dehn twists on the A-model side. Our treatment follows [AL15] and uses stable ∞ -categories to make the fiber and cofiber constructions canonical. The independent discussion of $\mathbb{P}^n$ -functors appears in an appendix.

1.1 Adjunctions and spherical twists

We begin with several standard facts about adjunctions of ∞ -categories; see [Lur26, Tag 02C9].

Definition 1.1.1. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ be functors of ∞ -categories. We say that F is left adjoint to G if there exist a unit $\eta: \mathrm{id}_{\mathcal{C}} \rightarrow GF$ and a counit $\varepsilon: FG \rightarrow \mathrm{id}_{\mathcal{D}}$ satisfying the triangle identities:

- The composite

$$F \xrightarrow{F\eta} FGF \xrightarrow{\varepsilon F} F$$

is id_F in the ∞ -category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$.

- The composite

$$G \xrightarrow{\eta G} GFG \xrightarrow{G\varepsilon} G$$

is id_G in the ∞ -category $\text{Fun}(\mathcal{D}, \mathcal{C})$.

Spherical adjunction.

Definition 1.1.2. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of stable ∞ -categories with left and right adjoints $L, R: \mathcal{D} \rightarrow \mathcal{C}$. The four associated functors are

$$\begin{aligned} T_{\mathcal{C}} &= \text{cofib}(\text{id}_{\mathcal{C}} \rightarrow RF), & T_{\mathcal{D}} &= \text{fib}(FR \rightarrow \text{id}_{\mathcal{D}}), \\ T'_{\mathcal{C}} &= \text{fib}(LF \rightarrow \text{id}_{\mathcal{C}}), & T'_{\mathcal{D}} &= \text{cofib}(\text{id}_{\mathcal{D}} \rightarrow FL). \end{aligned}$$

For $L \dashv F \dashv R$, write

$$\eta_L: \text{id}_{\mathcal{D}} \rightarrow FL, \quad \varepsilon_L: LF \rightarrow \text{id}_{\mathcal{C}}$$

for the unit and counit of $L \dashv F$, and

$$\eta_R: \text{id}_{\mathcal{C}} \rightarrow RF, \quad \varepsilon_R: FR \rightarrow \text{id}_{\mathcal{D}}$$

for those of $F \dashv R$.

Definition 1.1.3. An adjoint triple

$$L \dashv F \dashv R$$

between stable ∞ -categories is a *spherical adjunction* if the source twist $T_{\mathcal{C}}$ and the target cotwist $T_{\mathcal{D}}$ are equivalences. In this case, we also call F a *spherical functor*.

Lemma 1.1.4. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ have right adjoint R . There are natural equivalences

$$FT_{\mathcal{C}} \simeq T_{\mathcal{D}}F, \quad T_{\mathcal{C}}R \simeq RT_{\mathcal{D}}.$$

Proof. It suffices to prove $FT_{\mathcal{C}} \simeq T_{\mathcal{D}}F$; the other equivalence is analogous. The defining fiber and cofiber sequences form the pushout diagram

$$\begin{array}{ccccc} F \circ \text{id}_{\mathcal{C}} & \longrightarrow & FRF & \longrightarrow & \text{id}_{\mathcal{D}} \circ F \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & F \circ T_{\mathcal{C}} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow \\ & & 0 & \longrightarrow & T_{\mathcal{D}} \circ F[1] \end{array},$$

where 0 is the zero object and each square is a pushout. We therefore obtain the natural equivalence

$$FT_{\mathcal{C}} \simeq T_{\mathcal{D}}F.$$

□

Lemma 1.1.5. *For an adjoint triple $L \dashv F \dashv R$, there are adjunctions*

$$T'_{\mathcal{D}} \dashv T_{\mathcal{D}}, \quad T'_{\mathcal{C}} \dashv T_{\mathcal{C}}.$$

Proof. It suffices to prove $T'_{\mathcal{D}} \dashv T_{\mathcal{D}}$; the other adjunction is analogous. By the mapping-space criterion for adjunctions [Lur26, Corollary 6.2.6.5], it suffices to construct a natural equivalence

$$\mathrm{Map}_{\mathcal{D}}(M, T_{\mathcal{D}}N) \simeq \mathrm{Map}_{\mathcal{D}}(T'_{\mathcal{D}}M, N)$$

for all $M, N \in \mathcal{D}$. The defining fiber and cofiber sequences give

$$\begin{aligned} \mathrm{Map}_{\mathcal{D}}(M, T_{\mathcal{D}}N) &\simeq \mathrm{fib}(\mathrm{Map}_{\mathcal{D}}(M, FR(N)) \rightarrow \mathrm{Map}_{\mathcal{D}}(M, N)) \\ &\simeq \mathrm{fib}(\mathrm{Map}_{\mathcal{C}}(L(M), R(N)) \rightarrow \mathrm{Map}_{\mathcal{D}}(M, N)) \\ &\simeq \mathrm{fib}(\mathrm{Map}_{\mathcal{D}}(FL(M), N) \rightarrow \mathrm{Map}_{\mathcal{D}}(M, N)) \\ &\simeq \mathrm{Map}_{\mathcal{D}}(T'_{\mathcal{D}}M, N). \end{aligned}$$

Here postcomposition with $\varepsilon_R(N)$ corresponds, under the two adjunction equivalences, to precomposition with $\eta_L(M)$. □

We next construct the unit and counit of $T'_{\mathcal{D}} \dashv T_{\mathcal{D}}$. Write the defining sequences as

$$\mathrm{id}_{\mathcal{D}} \xrightarrow{\eta_L} FL \xrightarrow{q} T'_{\mathcal{D}}, \quad T_{\mathcal{D}} \xrightarrow{i} FR \xrightarrow{\varepsilon_R} \mathrm{id}_{\mathcal{D}}.$$

Since $\mathcal{D}$ is stable, the first sequence identifies $\mathrm{id}_{\mathcal{D}}$ with the fiber of q , and the second identifies $T_{\mathcal{D}}$ with the fiber of ε_R .

Constructing the unit. To construct the unit, it is enough to compare the arrow $q: FL \rightarrow T'_{\mathcal{D}}$ with $\varepsilon_R T'_{\mathcal{D}}: FRT'_{\mathcal{D}} \rightarrow T'_{\mathcal{D}}$. Define

$$f: FL \xrightarrow{F\eta_R L} FRFL \xrightarrow{FRq} FRT'_{\mathcal{D}}.$$

The identity $\varepsilon_R T'_\mathcal{D} \circ f = q$, proved below, gives a morphism of fiber sequences

$$\begin{array}{ccccc} \mathrm{id}_\mathcal{D} & \xrightarrow{\eta_L} & FL & \xrightarrow{q} & T'_\mathcal{D} \\ \exists \eta \downarrow & & \downarrow f & & \downarrow \mathrm{id} \\ T_\mathcal{D} T'_\mathcal{D} & \xrightarrow{i_{T'_\mathcal{D}}} & FR T'_\mathcal{D} & \xrightarrow{\varepsilon_R T'_\mathcal{D}} & T'_\mathcal{D} \end{array}$$

and the induced map on fibers is the unit $\eta: \mathrm{id}_\mathcal{D} \rightarrow T_\mathcal{D} T'_\mathcal{D}$.

Naturality of $\varepsilon_R: FR \rightarrow \mathrm{id}_\mathcal{D}$ with respect to $q: FL \rightarrow T'_\mathcal{D}$ gives the commutative square

$$\begin{array}{ccc} FRFL & \xrightarrow{FRq} & FR T'_\mathcal{D} \\ \varepsilon_R FL \downarrow & & \downarrow \varepsilon_R T'_\mathcal{D} \\ FL & \xrightarrow{q} & T'_\mathcal{D} \end{array}$$

Therefore

$$\varepsilon_R T'_\mathcal{D} \circ f = q \circ \varepsilon_R FL \circ F\eta_R L.$$

The last two factors are the triangle identity $\varepsilon_R F \circ F\eta_R = \mathrm{id}_F$ for $F \dashv R$, whiskered by L . Hence

$$\varepsilon_R FL \circ F\eta_R L = \mathrm{id}_{FL},$$

and therefore $\varepsilon_R T'_\mathcal{D} \circ f = q$.

Constructing the counit. For the counit, we compare the cofiber sequence defining $T'_\mathcal{D} T_\mathcal{D}$ with the sequence defining $\mathrm{id}_\mathcal{D}$ as the cofiber of i . Define

$$g: FL T_\mathcal{D} \xrightarrow{FLi} FLFR \xrightarrow{F\varepsilon_L R} FR.$$

The identity $g \circ \eta_L T_\mathcal{D} = i$ makes the left square below commute, so the universal property of the cofiber gives the counit $\varepsilon: T'_\mathcal{D} T_\mathcal{D} \rightarrow \mathrm{id}_\mathcal{D}$:

$$\begin{array}{ccccc} T_\mathcal{D} & \xrightarrow{\eta_L T_\mathcal{D}} & FL T_\mathcal{D} & \xrightarrow{q T_\mathcal{D}} & T'_\mathcal{D} T_\mathcal{D} \\ \mathrm{id} \downarrow & & \downarrow g & & \downarrow \exists \varepsilon \\ T_\mathcal{D} & \xrightarrow{i} & FR & \xrightarrow{\varepsilon_R} & \mathrm{id}_\mathcal{D} \end{array}$$

Indeed, naturality of η_L with respect to $i: T_\mathcal{D} \rightarrow FR$ gives

$$FLi \circ \eta_L T_\mathcal{D} = \eta_L FR \circ i.$$

Thus

$$g \circ \eta_L T_{\mathcal{D}} = F\varepsilon_L R \circ \eta_L F R \circ i = i,$$

where the last equality is the triangle identity $F\varepsilon_L \circ \eta_L F = \text{id}_F$ for $L \dashv F$, whiskered by R .

Theorem 1.1.6. *The maps η and ε satisfy the triangle identities and define the adjunction $T'_{\mathcal{D}} \dashv T_{\mathcal{D}}$.*

Proof. We verify the first triangle identity; the second follows from the dual argument in $\mathcal{D}^{\text{op}}$. Stack the unit construction, whiskered by $T_{\mathcal{D}}$, above the counit construction:

$$\begin{array}{ccccc} T_{\mathcal{D}} & \longrightarrow & FLT_{\mathcal{D}} & \longrightarrow & T'_{\mathcal{D}}T_{\mathcal{D}} \\ \eta T_{\mathcal{D}} \downarrow & & \downarrow fT_{\mathcal{D}} & & \downarrow \text{id} \\ T_{\mathcal{D}}T'_{\mathcal{D}}T_{\mathcal{D}} & \longrightarrow & FRT'_{\mathcal{D}}T_{\mathcal{D}} & \longrightarrow & T'_{\mathcal{D}}T_{\mathcal{D}} \\ T_{\mathcal{D}}\varepsilon \downarrow & & \downarrow FR\varepsilon & & \downarrow \varepsilon \\ T_{\mathcal{D}} & \longrightarrow & FR & \longrightarrow & \text{id}_{\mathcal{D}} \end{array}$$

It suffices to identify the composite in the middle column. Using $f = FRq \circ F\eta_R L$, we obtain

$$FR\varepsilon \circ fT_{\mathcal{D}} = FR(\varepsilon \circ qT_{\mathcal{D}}) \circ F\eta_R LT_{\mathcal{D}}.$$

By the square defining the counit, $\varepsilon \circ qT_{\mathcal{D}} = \varepsilon_R \circ g$, so

$$FR\varepsilon \circ fT_{\mathcal{D}} = FR\varepsilon_R \circ FRg \circ F\eta_R LT_{\mathcal{D}}.$$

Write $g = Fh$, where $h = \varepsilon_L R \circ Li: LT_{\mathcal{D}} \rightarrow R$. Naturality of η_R with respect to h gives

$$FRg \circ F\eta_R LT_{\mathcal{D}} = F(\eta_R R \circ h).$$

Therefore

$$FR\varepsilon \circ fT_{\mathcal{D}} = F(R\varepsilon_R \circ \eta_R R) \circ g = g,$$

by the triangle identity for $F \dashv R$. Thus the middle column is the map g used to define the counit. The universal property of the fiber defining $T_{\mathcal{D}}$ identifies the left column with $\text{id}_{T_{\mathcal{D}}}$. Hence

$$(T_{\mathcal{D}}\varepsilon) \circ (\eta T_{\mathcal{D}}) = \text{id}_{T_{\mathcal{D}}}.$$

□

1.2 The 2-out-of-4 criterion

The preceding adjunction yields the following criterion for sphericity.

Theorem 1.2.1 (2-out-of-4). *Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of stable ∞ -categories with left and right adjoints L and R . If any two of the following conditions hold, then all four hold:*

1. *The target cotwist $T_{\mathcal{D}}$ is an equivalence, or equivalently, the target dual twist $T'_{\mathcal{D}}$ is an equivalence.*
2. *The source twist $T_{\mathcal{C}}$ is an equivalence, or equivalently, the source dual cotwist $T'_{\mathcal{C}}$ is an equivalence.*
3. *The comparison map*

$$R \longrightarrow RFL \longrightarrow T_{\mathcal{C}}L$$

is an equivalence, or equivalently, so is

$$L \longrightarrow RFL \longrightarrow RT'_{\mathcal{D}}.$$

4. *The comparison map*

$$LT_{\mathcal{D}} \longrightarrow LFR \longrightarrow R$$

is an equivalence, or equivalently, so is

$$T'_{\mathcal{C}}R \longrightarrow LFR \longrightarrow L.$$

The following two lemmas identify the cofibers of the comparison maps in conditions (3) and (4).

Lemma 1.2.2. *Set $K = \text{cofib}(R \rightarrow T_{\mathcal{C}}L)$. There are natural equivalences*

$$\begin{aligned} K &\simeq \text{cofib}(L \rightarrow RT'_{\mathcal{D}}), \\ FK &\simeq \text{cofib}(\text{id}_{\mathcal{D}} \rightarrow T_{\mathcal{D}}T'_{\mathcal{D}}), \\ KF &\simeq \text{cofib}(\text{id}_{\mathcal{C}} \rightarrow T_{\mathcal{C}}T'_{\mathcal{C}}). \end{aligned}$$

Proof. We directly compare the two maps with common target RFL :

$$L \xrightarrow{\eta_R L} RFL \longrightarrow T_{\mathcal{C}}L, \quad R \xrightarrow{R\eta_L} RFL \longrightarrow RT'_{\mathcal{D}}.$$

Filling in the induced cofibers gives the diagram

$$\begin{array}{ccccc}
 R & \longrightarrow & RFL & \longrightarrow & T_c L \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & RT'_D & \longrightarrow & K \\
 & & \downarrow & & \downarrow \\
 & & 0 & \longrightarrow & L[1]
 \end{array}$$

whose small squares are pushouts. The right-hand part gives $K \simeq \text{cofib}(L \rightarrow RT'_D)$.

Next apply F to the defining cofiber sequence of K . Using the map

$$f : FL \xrightarrow{F\eta_R L} FRFL \xrightarrow{FRq} FRT'_D$$

from the construction of the unit, the diagram induced by these maps is

$$\begin{array}{ccccc}
 FL & \longrightarrow & FRT'_D & \longrightarrow & T'_D \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & FK & \longrightarrow & \text{id}_D[1] \\
 & & \downarrow & & \downarrow \\
 & & 0 & \longrightarrow & T_D T'_D[1]
 \end{array}$$

Rotating the right column gives

$$FK \simeq \text{cofib}(\text{id}_D \rightarrow T_D T'_D).$$

The third equivalence follows in the same way after composing with F on the right. The relevant diagram is

$$\begin{array}{ccccc}
 RF & \longrightarrow & T_c LF & \longrightarrow & T_c \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & KF & \longrightarrow & \text{id}_c[1] \\
 & & \downarrow & & \downarrow \\
 & & 0 & \longrightarrow & T_c T'_c[1]
 \end{array}$$

and gives

$$KF \simeq \text{cofib}(\text{id}_c \rightarrow T_c T'_c).$$

□

Lemma 1.2.3. *Set $K' = \text{cofib}(LT_{\mathcal{D}} \rightarrow R)$. There is a natural equivalence*

$$K' \simeq \text{cofib}(T'_{\mathcal{C}}R \rightarrow L).$$

If $A \in \mathcal{D}$ and $T_{\mathcal{D}}(A) = 0$, then $A \simeq FK'(A)$. Similarly, if $B \in \mathcal{C}$ and $T_{\mathcal{C}}(B) = 0$, then $B \simeq K'F(B)$.

Proof. Start with the map

$$\alpha : LT_{\mathcal{D}} \xrightarrow{Li} LFR \xrightarrow{\varepsilon_L R} R,$$

whose cofiber is K' . We compare it with the two defining cofiber sequences

$$LT_{\mathcal{D}} \xrightarrow{Li} LFR \rightarrow L, \quad T'_{\mathcal{C}}R \rightarrow LFR \xrightarrow{\varepsilon_L R} R,$$

and obtain the diagram

$$\begin{array}{ccccc} LT_{\mathcal{D}} & \longrightarrow & LFR & \longrightarrow & R \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & L & \longrightarrow & K' \\ & & \downarrow & & \downarrow \\ & & 0 & \longrightarrow & T'_{\mathcal{C}}R[1] \end{array}$$

Thus the right column gives $K' \simeq \text{cofib}(T'_{\mathcal{C}}R \rightarrow L)$.

Now let $A \in \mathcal{D}$ and suppose $T_{\mathcal{D}}(A) = 0$. The defining fiber sequence

$$T_{\mathcal{D}}(A) \rightarrow FR(A) \xrightarrow{\varepsilon_R(A)} A$$

shows that $FR(A) \simeq A$. Since the first term of

$$LT_{\mathcal{D}}(A) \rightarrow R(A) \rightarrow K'(A)$$

vanishes, we also have $R(A) \simeq K'(A)$. Therefore $A \simeq FK'(A)$.

Similarly, if $T_{\mathcal{C}}(B) = 0$, then the unit $B \rightarrow RF(B)$ is an equivalence. Moreover, the equivalence $T_{\mathcal{D}}F \simeq FT_{\mathcal{C}}$ shows that the first term in

$$LT_{\mathcal{D}}F(B) \rightarrow RF(B) \rightarrow K'F(B)$$

vanishes. Hence $K'F(B) \simeq RF(B) \simeq B$. □

Proposition 1.2.4. *If either condition (1) or condition (2) of Theorem 1.2.1 holds, then condition (3) is equivalent to condition (4).*

Proof. Assume first that condition (1) holds, so $T_{\mathcal{D}}$ and $T'_{\mathcal{D}}$ are equivalences. Consider the two natural maps

$$\alpha : LT_{\mathcal{D}} \xrightarrow{Li} LFR \xrightarrow{\varepsilon_L R} R, \quad \beta : L \xrightarrow{\eta_R L} RFL \xrightarrow{Rq} RT'_{\mathcal{D}}.$$

Here β is the second formulation of condition (3), while α is the first formulation of condition (4). We show that they are identified after composing with $T'_{\mathcal{D}}$.

From the construction of the unit $\eta : \text{id}_{\mathcal{D}} \rightarrow T_{\mathcal{D}}T'_{\mathcal{D}}$, we have

$$iT'_{\mathcal{D}} \circ \eta = f \circ \eta_L.$$

Since $f = FRq \circ F\eta_R L = F\beta$, this becomes

$$iT'_{\mathcal{D}} \circ \eta = F\beta \circ \eta_L.$$

Now compute:

$$\begin{aligned} \alpha T'_{\mathcal{D}} \circ L\eta &= \varepsilon_L RT'_{\mathcal{D}} \circ LiT'_{\mathcal{D}} \circ L\eta \\ &= \varepsilon_L RT'_{\mathcal{D}} \circ L(iT'_{\mathcal{D}} \circ \eta) \\ &= \varepsilon_L RT'_{\mathcal{D}} \circ L(F\beta \circ \eta_L) \\ &= \varepsilon_L RT'_{\mathcal{D}} \circ LF\beta \circ L\eta_L \\ &= \beta \circ \varepsilon_L L \circ L\eta_L \\ &= \beta. \end{aligned}$$

The fifth equality is naturality of $\varepsilon_L : LF \rightarrow \text{id}_{\mathcal{C}}$, applied to $\beta : L \rightarrow RT'_{\mathcal{D}}$; the final equality is the triangle identity for $L \dashv F$. Since $T'_{\mathcal{D}}$ and $L\eta$ are equivalences, α is an equivalence if and only if β is. Thus condition (3) is equivalent to condition (4) whenever condition (1) holds.

If condition (2) holds instead, apply the same argument to $T'_{\mathcal{C}} \dashv T_{\mathcal{C}}$. It identifies the two alternative comparison maps in conditions (3) and (4). The two conditions are therefore equivalent whenever either condition (1) or condition (2) holds. $\square$

Proof of Theorem 1.2.1. Proposition 1.2.4 reduces the mixed cases to

$$(1) + (2) \implies (3) + (4), \quad (3) + (4) \implies (1) + (2).$$

Assume first that (3) and (4) hold. Then $K \simeq 0$ and $K' \simeq 0$. The first lemma identifies the unit maps

$$\mathrm{id}_{\mathcal{D}} \rightarrow T_{\mathcal{D}}T'_{\mathcal{D}}, \quad \mathrm{id}_{\mathcal{C}} \rightarrow T_{\mathcal{C}}T'_{\mathcal{C}}$$

as equivalences, so $T'_{\mathcal{D}}$ and $T'_{\mathcal{C}}$ are fully faithful. The second lemma shows that $T_{\mathcal{D}}$ and $T_{\mathcal{C}}$ are conservative: if $T_{\mathcal{D}}(A) = 0$, then $A \simeq FK'(A) = 0$, and similarly on $\mathcal{C}$. Thus $T'_{\mathcal{D}} \dashv T_{\mathcal{D}}$ and $T'_{\mathcal{C}} \dashv T_{\mathcal{C}}$ are adjoint equivalences. Hence (1) and (2) hold.

Conversely, assume (1) and (2). The left adjoints $T'_{\mathcal{D}}$ and $T'_{\mathcal{C}}$ are then the respective inverse equivalences, and the first lemma gives

$$FK \simeq 0, \quad KF \simeq 0.$$

It remains to show that $K \simeq 0$. The unit $L \rightarrow LFL$ is a section of the counit $LFL \rightarrow L$ by the triangle identity. Hence the fiber of the counit is canonically equivalent to the cofiber of the unit; in the present notation,

$$T'_{\mathcal{C}}L \simeq LT'_{\mathcal{D}}.$$

Equivalently, the two split cofiber sequences

$$T'_{\mathcal{C}}L \rightarrow LFL \rightarrow L, \quad L \rightarrow LFL \rightarrow LT'_{\mathcal{D}}$$

arise from the same canonical splitting and give

$$LFL \simeq L \oplus T'_{\mathcal{C}}L \simeq L \oplus LT'_{\mathcal{D}}.$$

Let

$$T_{\mathcal{C}}L \xrightarrow{\sim} LT_{\mathcal{D}}$$

be the inverse of the double mate of this canonical equivalence. Under this identification, the composite

$$R \xrightarrow{R\eta_L} RFL \rightarrow T_{\mathcal{C}}L \simeq LT_{\mathcal{D}} \xrightarrow{Li} LFR \xrightarrow{\varepsilon_L R} R$$

is the identity. Indeed, compatibility of mates with composition identifies its successive mate under $L \dashv F \dashv R$ with

$$L \xrightarrow{L\eta_L} LFL \xrightarrow{\varepsilon_L L} L,$$

which is id_L by the triangle identity. Hence the cofiber sequence

$$R \rightarrow T_{\mathcal{C}}L \rightarrow K$$

splits, so $T_{\mathcal{C}}L \simeq R \oplus K$.

For any $X \in \mathcal{D}$, we then have

$$\begin{aligned} \text{Map}_{\mathcal{C}}(T_{\mathcal{C}}L(X), K(X)) &\simeq \text{Map}_{\mathcal{C}}(LT_{\mathcal{D}}(X), K(X)) \\ &\simeq \text{Map}_{\mathcal{D}}(T_{\mathcal{D}}(X), FK(X)) \\ &\simeq 0. \end{aligned}$$

Since $K(X)$ is a direct summand of $T_{\mathcal{C}}L(X)$, this forces $K(X) \simeq 0$. Thus $K \simeq 0$, so (3) holds. Finally, (4) follows from Proposition 1.2.4 because $T_{\mathcal{D}}$ is an equivalence. $\square$

1.3 Monadic constructions

Spherical functors are a useful source of autoequivalences of stable ∞ -categories in geometry. Conversely, every autoequivalence of a stable ∞ -category can be realized as the twist of a spherical functor. The following criterion is the monadic sphericity theorem of [Chr22].

Theorem 1.3.1. *Let $\mathcal{C}$ be a stable ∞ -category, and let T be a monad on $\mathcal{C}$ with unit $t: \text{id}_{\mathcal{C}} \rightarrow T$. Set*

$$T_{\mathcal{C}} = \text{cofib}(\text{id}_{\mathcal{C}} \xrightarrow{t} T).$$

The following conditions are equivalent:

1. *The functor $T_{\mathcal{C}}$ is an equivalence and $T_{\mathcal{C}}t \simeq tT_{\mathcal{C}}$.*
2. *The free–forgetful adjunction*

$$F: \mathcal{C} \rightleftarrows \text{LMod}_T(\mathcal{C}) : G$$

is spherical.

Corollary 1.3.2. *Every autoequivalence of a stable ∞ -category is the twist of a spherical adjunction.*

Proof. For an autoequivalence T of $\mathcal{C}$, the functor $M = \text{id}_{\mathcal{C}} \oplus T$ has a square-zero extension monad structure. The theorem applies to the free–forgetful adjunction for M , whose source twist is T . $\square$

1.4 Examples

We conclude with several geometric examples. We first record a practical criterion. In this section, $D(X)$ means $D_{\text{qc}}(X)$.

Theorem 1.4.1. *Let X and Y be perfect stacks, and set $\mathcal{C} = D(X)$ and $\mathcal{D} = D(Y)$. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a Fourier–Mukai functor represented by a perfect kernel. If $T_{\mathcal{C}}$ is an autoequivalence, then any equivalence $R \simeq T_{\mathcal{C}}L$ implies that the canonical map in condition (3) of Theorem 1.2.1 is an equivalence.*

Proof. Use the spanning class $\Omega = \text{im}(F) \cup \ker R$. It is right spanning. Indeed, suppose that $M \in \mathcal{D}$ is right orthogonal to every object of Ω . Orthogonality to $\text{im}(F)$ gives $R(M) = 0$, so $M \in \ker R \subset \Omega$; testing against M itself gives $M = 0$.

The equivalence of $T_{\mathcal{C}}$ implies $KF = 0$. For an object $M \in \ker R$, consider the fiber sequence

$$R(M) \longrightarrow T_{\mathcal{C}}L(M) \longrightarrow K(M).$$

The equivalence $R \simeq T_{\mathcal{C}}L$ implies $R(M) \simeq T_{\mathcal{C}}L(M) \simeq 0$, so $K(M) \simeq 0$. Thus K vanishes on Ω . By the assumptions, K is again a Fourier–Mukai functor represented by a perfect kernel, and hence admits a right adjoint K^R . For every $N \in \mathcal{C}$ and $M \in \Omega$, adjunction gives

$$\text{Map}_{\mathcal{D}}(M, K^R(N)) \simeq \text{Map}_{\mathcal{C}}(K(M), N) \simeq 0.$$

Since Ω is right spanning, $K^R(N) = 0$ for every N , and hence $K = 0$. Therefore the canonical comparison map in condition (3) is an equivalence. $\square$

1. Classical spherical objects.

Let X be a smooth proper variety of dimension n over a field k . A perfect complex $\mathcal{E} \in D(X)$ is spherical if

$$\text{RHom}_X(\mathcal{E}, \mathcal{E}) \simeq H^*(S^n, \mathbb{Z}) \otimes_{\mathbb{Z}} k \simeq k \oplus k[-n], \quad \mathcal{E} \otimes \omega_X \simeq \mathcal{E}.$$

Consider

$$F = \mathcal{E} \otimes -: D(k) \longrightarrow D(X).$$

Its right adjoint is $R = \text{RHom}_X(\mathcal{E}, -)$, and

$$RF \simeq \text{RHom}_X(\mathcal{E}, \mathcal{E}) \otimes - \simeq \text{id} \oplus [-n].$$

Hence $T_{\mathcal{C}} \simeq [-n]$. By the Serre functor $S_k \simeq \text{id}$ and $S_X \simeq (-) \otimes \omega_X[n]$, the condition $R \simeq T_{\mathcal{C}}L$ is equivalent to

$$S_X F T_{\mathcal{C}}(-) \simeq (\mathcal{E} \otimes -[-n]) \otimes \omega_X[n] \simeq (\mathcal{E} \otimes \omega_X) \otimes - \simeq F S_k(-).$$

Thus F is spherical by Theorem 1.2.1; its usual twist is the Seidel–Thomas twist [ST01].

2. EZ -spherical objects.

Let $i: E \hookrightarrow X$ be a regular embedding of codimension d , and let $q: E \rightarrow Z$ be flat of relative dimension k , where X, E, Z are smooth and E is proper. Assume that a line bundle θ on Z satisfies

$$q^*(\theta \otimes \omega_Z^{-1}) \simeq i^* \omega_X^{-1}.$$

Write $\nu = N_{E/X}$. A perfect complex $K \in D(E)$ is EZ -spherical if, for $d > 0$,

$$q_* \text{RHom}_E(K, K) \simeq \mathcal{O}_Z, \quad q_* \text{RHom}_E(K, K \otimes \Lambda^c \nu) \simeq 0 \quad (0 < c < d),$$

while for $d = 0$ one requires a fiber sequence

$$\mathcal{O}_Z \longrightarrow q_* \text{RHom}_E(K, K) \longrightarrow \theta[-k].$$

Then

$$F_K: D(Z) \longrightarrow D(X), \quad M \longmapsto i_*(K \otimes q^* M),$$

is spherical [Hor05, Definition 2.1 and Theorem 2.11].

3. Toda's fat spherical objects.

Let X be a smooth projective complex variety of dimension n , A a local Artinian $\mathbb{C}$ -algebra, and $\pi: X \times \text{Spec } A \rightarrow X$ the projection. Let $K \in D(X \times \text{Spec } A)$ be represented by a bounded complex of A -flat coherent sheaves, and set $E = K|_{X \times \{0\}}$. The complex K is A -spherical if

$$\text{Ext}_X^i(\pi_* K, E) \simeq \begin{cases} \mathbb{C}, & i = 0, n, \\ 0, & \text{otherwise,} \end{cases} \quad E \otimes \omega_X \simeq E.$$

Consider the Fourier–Mukai functor

$$F_K: D(\text{Spec } A) \longrightarrow D(X), \quad M \longmapsto \pi_*(K \otimes \pi'^* M),$$

where $\pi': X \times \text{Spec } A \rightarrow \text{Spec } A$ is the second projection. The functor F_K is spherical, and its twist is an autoequivalence [Tod07, Definition 2.1 and Theorem 2.3].

4. *Hilbert schemes of points on K3 surfaces.*

Let S be a complex K3 surface, and let $S^{[n]}$ be its Hilbert scheme of length n subschemes. Write $Z \subset S \times S^{[n]}$ for the universal subscheme, with ideal sheaf $\mathcal{I}_Z$, and let

$$F: D(S) \longrightarrow D(S^{[n]})$$

be the Fourier–Mukai transform with kernel $\mathcal{I}_Z$. Then

$$RF \simeq \mathrm{id}_S \oplus [-2] \oplus \cdots \oplus [2 - 2n].$$

For $n = 2$, the functor F is spherical. For $n > 2$, the same calculation supplies the motivating example of a $\mathbb{P}^{n-1}$ -functor rather than a spherical functor; see [Add16, Theorem 2].

5. *Noncommutative K3 surfaces.*

Let $X \subset \mathbb{P}^5$ be a smooth cubic hypersurface, and let

$$\mathcal{A} := \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle^\perp \subset D(X)$$

be Kuznetsov’s K3 subcategory. Let Y be the variety of lines on X .

More precisely, let

$$\mathcal{L} = \{(x, \ell) \in X \times Y \mid x \in \ell\}$$

be the universal line, with projections $X \xleftarrow{q} \mathcal{L} \xrightarrow{p} Y$. If $I: \mathcal{A} \hookrightarrow D(X)$ denotes the inclusion, then the functor

$$F = p_* q^* I: \mathcal{A} \longrightarrow D(Y)$$

satisfies $RF \simeq \mathrm{id}_{\mathcal{A}} \oplus [-2]$ and is spherical; see [Add16, Theorem 4].

Appendix A

Notes on $\mathbb{P}^n$ -functors

Status. This appendix records the construction of $\mathbb{P}^n$ -functors and their twists. It is not used elsewhere in the manuscript. The final autoequivalence theorem is quoted from Anno–Logvinenko; its direct proof in the present language is omitted.

Spherical functors provide a categorical framework for Dehn twists and their mirror-symmetric analogues. Their higher-dimensional variants are $\mathbb{P}^n$ -functors, introduced as functorial analogues of $\mathbb{P}^n$ -objects. Roughly, a functor

$$F: \mathcal{C} \longrightarrow \mathcal{D}$$

is a $\mathbb{P}^n$ -functor if its adjunction monad behaves like the cohomology algebra of projective space:

$$RF \simeq \mathrm{id}_{\mathcal{C}} \oplus H \oplus H^2 \oplus \cdots \oplus H^n,$$

where R is a right adjoint of F , H is an autoequivalence of $\mathcal{C}$, and multiplication is compatible with the algebra structure of

$$H^*(\mathbb{P}^n; k) \simeq k[h]/(h^{n+1}).$$

Such a functor determines a twist autoequivalence of $\mathcal{D}$, which recovers the spherical twist when $n = 1$.

A.1 Coherent complexes and convolution

The definition of a $\mathbb{P}^n$ -twist uses the convolution of a complex. Given a finite complex in a stable ∞ -category,

$$A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_m,$$

one would like to take its “total object”, obtained by iteratively forming cofibers. A sequence in the homotopy category alone does not determine such an object canonically: different choices of null-homotopies and higher coherences may lead to nonisomorphic convolutions. A stable ∞ -category retains precisely the coherent data needed to make these choices explicit.

The following results are standard; see [Ari21] for complete proofs.

Let $\mathcal{C}$ be a pointed ∞ -category. Let Ξ denote the pointed category whose objects are the integers together with a distinguished base point $*$, which is both initial and final. For integers $m, n \in \mathbb{Z}$, the morphisms in Ξ are given by

$$\mathrm{Hom}_{\Xi}(m, n) = \begin{cases} \{*\}, & n \neq m, \ m-1, \\ \{\mathrm{id}_m, *\}, & n = m, \\ \{\partial, *\}, & n = m-1. \end{cases}$$

Here $*$ denotes the zero morphism, and the composition rule implies $\partial \circ \partial = *$ whenever this expression is defined.

Definition A.1.1. A *coherent chain complex* in $\mathcal{C}$ is a pointed functor

$$X_{\bullet}: \Xi \longrightarrow \mathcal{C}.$$

We write

$$\mathrm{Ch}_{\bullet}(\mathcal{C}) := \mathrm{Fun}_{*}(\Xi, \mathcal{C})$$

for the ∞ -category of coherent chain complexes in $\mathcal{C}$.

Dually, a *coherent cochain complex* in $\mathcal{C}$ is a pointed functor

$$X^{\bullet}: \Xi^{\mathrm{op}} \longrightarrow \mathcal{C},$$

and we write

$$\mathrm{Ch}^{\bullet}(\mathcal{C}) := \mathrm{Fun}_{*}(\Xi^{\mathrm{op}}, \mathcal{C}).$$

Now assume that $\mathcal{C}$ is stable. Set

$$\mathrm{Fil}(\mathcal{C}) := \mathrm{Fun}(\mathbb{Z}^{\mathrm{op}}, \mathcal{C}), \quad \mathrm{gr}(\mathcal{C}) := \mathrm{Fun}(\mathbb{Z}_{\mathrm{disc}}, \mathcal{C}),$$

and let $X \in \mathrm{Fil}(\mathcal{C})$, with stages X^s . Its associated graded pieces are

$$\mathrm{gr}^s X := \mathrm{cofib}(X^{s+1} \rightarrow X^s).$$

For every s , the cofiber sequence

$$\mathrm{gr}^{s+1} X \longrightarrow X^s / X^{s+2} \longrightarrow \mathrm{gr}^s X$$

gives a connecting morphism

$$\partial_s: \mathrm{gr}^s X \longrightarrow \mathrm{gr}^{s+1} X[1].$$

The connecting morphisms assemble into the coherent cochain complex

$$\cdots \longrightarrow \mathrm{gr}^{-2} X[-2] \longrightarrow \mathrm{gr}^{-1} X[-1] \longrightarrow \mathrm{gr}^0 X \longrightarrow \mathrm{gr}^1 X[1] \longrightarrow \cdots.$$

We denote this coherent cochain complex by

$$\mathrm{gr}^\bullet X[\bullet] \in \mathrm{Ch}^\bullet(\mathcal{C}).$$

Theorem A.1.2 (Ariotta's E_1 -page theorem). *Let $\mathcal{C}$ be a stable ∞ -category with sequential limits. The associated graded functor*

$$\mathrm{gr}: \mathrm{Fil}(\mathcal{C}) \longrightarrow \mathrm{gr}(\mathcal{C})$$

factors through the forgetful functor

$$\mathrm{Ch}^\bullet(\mathcal{C}) \longrightarrow \mathrm{gr}(\mathcal{C})$$

and induces an equivalence

$$\mathrm{Fil}^{\mathrm{cpl}}(\mathcal{C}) \simeq \mathrm{Ch}^\bullet(\mathcal{C})$$

between the ∞ -category of complete decreasing filtrations in $\mathcal{C}$ and the ∞ -category of coherent cochain complexes in $\mathcal{C}$.

Here $\mathrm{Fil}^{\mathrm{cpl}}(\mathcal{C})$ denotes the full subcategory of complete decreasing filtrations. This equivalence provides a canonical definition of convolution.

Definition A.1.3. For a complex $C^\bullet \in \mathrm{Ch}^\bullet(\mathcal{C})$, define its convolution, or realization, by the composite

$$\mathrm{Ch}^\bullet(\mathcal{C}) \simeq \mathrm{Fil}^{\mathrm{cpl}}(\mathcal{C}) \xrightarrow{|\cdot|} \mathcal{C},$$

where $|X| := \mathrm{colim}_{s \in \mathbb{Z}^{\mathrm{op}}} X^s$.

A.1.1 Monoidal structure

From now on, assume that $\mathcal{C}$ is a stable monoidal category that admits sequential colimits and whose tensor product preserves colimits in each variable.

The addition map

$$+ : \mathbb{Z}^{\text{op}} \times \mathbb{Z}^{\text{op}} \longrightarrow \mathbb{Z}^{\text{op}}$$

endows $\text{Fil}(\mathcal{C})$ with the Day convolution monoidal structure. Namely, for $X, Y \in \text{Fil}(\mathcal{C})$, we define

$$X \otimes_{\text{Fil}} Y := +_!(X \boxtimes Y),$$

where $+_!$ denotes left Kan extension along $+$. Therefore

$$(X \otimes_{\text{Fil}} Y)^s \simeq \text{colim}_{p+q \geq s} X^p \otimes Y^q.$$

The inequality follows from the opposite order.

Lemma A.1.4. *The realization functor*

$$|-| : \text{Fil}(\mathcal{C}) \rightarrow \mathcal{C}$$

is monoidal. In other words, there is a canonical equivalence

$$|X \otimes_{\text{Fil}} Y| \simeq |X| \otimes |Y|.$$

Proof. By definition,

$$|X \otimes_{\text{Fil}} Y| \simeq \text{colim}_{s \in \mathbb{Z}^{\text{op}}} (X \otimes_{\text{Fil}} Y)^s.$$

Using the formula for Day convolution, this becomes

$$|X \otimes_{\text{Fil}} Y| \simeq \text{colim}_{s \in \mathbb{Z}^{\text{op}}} \text{colim}_{p+q \geq s} X^p \otimes Y^q.$$

Let

$$I := \{(p, q, s) \in \mathbb{Z}^3 \mid p + q \geq s\}$$

with the order induced from $(\mathbb{Z}^{\text{op}})^3$, and let

$$P := \mathbb{Z}^{\text{op}} \times \mathbb{Z}^{\text{op}}.$$

The above iterated colimit is the same as

$$\text{colim}_{(p,q,s) \in I} X^p \otimes Y^q.$$

Let

$$\pi : I \rightarrow P, \quad (p, q, s) \mapsto (p, q).$$

We claim that π is final. Fix $(a, b) \in P$. The category

$$I_{(a,b)/} := I \times_P P_{(a,b)/}$$

has objects triples (p, q, s) satisfying

$$a \geq p, \quad b \geq q, \quad p + q \geq s.$$

This category has an initial object, namely

$$(a, b, a + b).$$

Indeed, for every object $(p, q, s) \in I_{(a,b)/}$, we have

$$a \geq p, \quad b \geq q, \quad a + b \geq p + q \geq s,$$

so there is a morphism $(a, b, a + b) \rightarrow (p, q, s)$ in the induced opposite order. Thus $I_{(a,b)/}$ has contractible nerve. Therefore π is final, and hence

$$\operatorname{colim}_{(p,q,s) \in I} X^p \otimes Y^q \simeq \operatorname{colim}_{(p,q) \in P} X^p \otimes Y^q.$$

Since $\otimes$ preserves colimits separately,

$$\operatorname{colim}_{p,q} X^p \otimes Y^q \simeq \left(\operatorname{colim}_p X^p \right) \otimes \left(\operatorname{colim}_q Y^q \right).$$

Therefore

$$|X \otimes_{\text{Fil}} Y| \simeq |X| \otimes |Y|.$$

□

A.2 $\mathbb{P}^n$ -functors and their twists

Let

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

be an exact functor between stable ∞ -categories. Assume that F has both adjoints

$$L \dashv F \dashv R.$$

The monad RF is an algebra object in the monoidal category $\operatorname{End}(\mathcal{C})$.

Definition A.2.1. A $\mathbb{P}^n$ -structure on RF consists of

1. an autoequivalence $H \in \text{Aut}(\mathcal{C})$ such that $H^{n+1}(\ker F) = \ker F$;
2. an algebra object

$$M \in \text{Alg}(\text{Fil}(\text{End}(\mathcal{C}))).$$

Write $M_i := M^{-i}$ and $\text{gr}_i M := \text{gr}^{-i} M$. We require

$$M_i \simeq 0 \quad (i < 0), \quad M_0 \simeq \text{id}_{\mathcal{C}}, \quad M_i \simeq RF \quad (i \geq n),$$

and that the map

$$M_0 \longrightarrow M_n \simeq RF$$

corresponds to the unit of the monad

$$\eta : \text{id}_{\mathcal{C}} \longrightarrow RF.$$

3. equivalences

$$\text{gr}_i M \simeq H^i, \quad 0 \leq i \leq n,$$

compatible with composition $H^i \circ H^j \simeq H^{i+j}$, truncated in degrees $> n$;
all other graded pieces vanish;

4. an equivalence $R \simeq H^n L$.

Let

$$\varepsilon : FR \longrightarrow \text{id}_{\mathcal{D}}$$

be the counit of the adjunction $F \dashv R$. Consider the natural transformation

$$\gamma_F : FRFR \longrightarrow FR$$

defined by

$$\gamma_F := FR\varepsilon - \varepsilon FR.$$

Here the difference is taken in the stable ∞ -category $\text{End}(\mathcal{D})$.

Using the identification

$$FRFR \simeq F(RF)R,$$

we view γ_F as a morphism

$$FM_n R \longrightarrow FR.$$

By the unit compatibility above, the restriction of γ_F to the degree zero part

$$FM_0R \simeq FR$$

is null-homotopic. Indeed, it is the difference of the two maps

$$FR \xrightarrow{F\eta_R} FRFR \rightrightarrows FR,$$

both of which are homotopic to id_{FR} by the triangle identities.

Restrict γ_F along the structure map $FM_1R \rightarrow FM_nR$. Since its restriction to FM_0R is null-homotopic, the resulting map factors through the cofiber

$$\text{cofib}(FM_0R \rightarrow FM_1R) \simeq F \text{gr}_1 MR \simeq FHR.$$

We denote the induced morphism by

$$\psi : FHR \rightarrow FR.$$

Definition A.2.2. The $\mathbb{P}^n$ -twist associated to F is the endofunctor

$$T_F \in \text{End}(\mathcal{D})$$

defined as the convolution

$$T_F := |FHR \xrightarrow{\psi} FR \xrightarrow{\varepsilon} \text{id}_{\mathcal{D}}|.$$

The left adjoints of the three terms in the defining complex are

$$(FHR)^L \simeq FH^{-1}L, \quad (FR)^L \simeq FL, \quad (\text{id}_{\mathcal{D}})^L \simeq \text{id}_{\mathcal{D}}.$$

Taking left adjoints reverses the arrows. Hence the left adjoint of T_F is given by the realization

$$T_F^L \simeq |\text{id}_{\mathcal{D}} \rightarrow FL \rightarrow FH^{-1}L|,$$

where the first map is the unit of the adjunction $L \dashv F$, and the second map is the left adjoint of

$$\psi : FHR \rightarrow FR.$$

The Day convolution above induces a monoidal structure on coherent cochain complexes. Let $\mathcal{V}$ be a stable monoidal ∞ -category to which the previous discussion applies. Following [Ari21, Definition 6.7, Remark 6.8, Corollary 6.10], we write

$$\otimes_{\text{cc}}$$

for the tensor product on coherent cochain complexes induced from the completed Day convolution product on filtered objects. Thus, for bounded $K^\bullet, L^\bullet \in \mathbf{Ch}^\bullet(\mathcal{V})$, one has

$$(K^\bullet \otimes_{\text{cc}} L^\bullet)^m \simeq \bigoplus_{p+q=m} K^p \otimes L^q.$$

Moreover, realization is monoidal:

$$|K^\bullet \otimes_{\text{cc}} L^\bullet| \simeq |K^\bullet| \otimes |L^\bullet|.$$

When $\mathcal{V} = \mathbf{End}(\mathcal{D})$, the monoidal product is composition of endofunctors, so the formula becomes

$$(K^\bullet \otimes_{\text{cc}} L^\bullet)^m \simeq \bigoplus_{p+q=m} K^p \circ L^q.$$

Theorem A.2.3 (Anno–Logvinenko [AL25]). *The $\mathbb{P}^n$ -twist T_F is an autoequivalence of $\mathcal{D}$.*

A direct proof in the present language is omitted.

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